

1 Solution — GIAM 7.4

Problem 3

1.1 Problem

Find $(x^2 + y^2)^6$.

1.2 Setup — Binomial Theorem with a Substitution

The binomial theorem states

$$(u + v)^n = \sum_{k=0}^n \binom{n}{k} u^{n-k} v^k.$$

Take $u = x^2$, $v = y^2$, $n = 6$:

$$(x^2 + y^2)^6 = \sum_{k=0}^6 \binom{6}{k} (x^2)^{6-k} (y^2)^k = \sum_{k=0}^6 \binom{6}{k} x^{2(6-k)} y^{2k}.$$

The substitution $u = x^2, v = y^2$ has two consequences worth noting upfront:

- Every exponent of x and y in the answer is **even** — specifically, $2(6 - k)$ on x and $2k$ on y .
- Every term has *total* degree $2(6 - k) + 2k = 12$. The answer is a *homogeneous* polynomial of degree 12.

1.3 Step 1 — Pascal's Row 6

$$\binom{6}{0} = 1, \quad \binom{6}{1} = 6, \quad \binom{6}{2} = 15, \quad \binom{6}{3} = 20, \quad \binom{6}{4} = 15, \quad \binom{6}{5} = 6, \quad \binom{6}{6} = 1.$$

(Same coefficients as in Problem 1 — 1001⁶ used row 6 of Pascal as well.)

1.4 Step 2 — Compute Each Term

k	$\binom{6}{k}$	$x\text{-exponent} = 2(6 - k)$	$y\text{-exponent} = 2k$	Term
0	1	12	0	x^{12}
1	6	10	2	$6x^{10}y^2$
2	15	8	4	$15x^8y^4$
3	20	6	6	$20x^6y^6$
4	15	4	8	$15x^4y^8$
5	6	2	10	$6x^2y^{10}$
6	1	0	12	y^{12}

Table 1.1

1.5 Step 3 — Assemble

$$(x^2 + y^2)^6 = x^{12} + 6x^{10}y^2 + 15x^8y^4 + 20x^6y^6 + 15x^4y^8 + 6x^2y^{10} + y^{12}.$$

1.6 Sanity Checks

$y = 0$. $(x^2 + 0)^6 = x^{12}$. Polynomial reduces to x^{12} . ✓

$x = 0$. $(0 + y^2)^6 = y^{12}$. Polynomial reduces to y^{12} . ✓

$x = y = 1$. $(1 + 1)^6 = 64$. Polynomial at $(1, 1)$:

$$1 + 6 + 15 + 20 + 15 + 6 + 1 = 64. \checkmark$$

$x \leftrightarrow y$ **symmetry**. Swapping x and y leaves $x^2 + y^2$ unchanged, so it must leave the expansion unchanged. Reading the answer's monomials in order — x^{12} , $6x^{10}y^2$, $15x^8y^4$, $20x^6y^6$, $15x^4y^8$, $6x^2y^{10}$, y^{12} — the pairs swap into each other:

$$x^{12} \leftrightarrow y^{12}, \quad 6x^{10}y^2 \leftrightarrow 6x^2y^{10}, \quad 15x^8y^4 \leftrightarrow 15x^4y^8,$$

and the middle term $20x^6y^6$ is fixed. The expansion is *invariant* under $x \leftrightarrow y$, as required. ✓

1.7 Remark — Re-grouping the Answer by Symmetry

Pairing terms that are swapped by $x \leftrightarrow y$:

$$(x^2 + y^2)^6 = (x^{12} + y^{12}) + 6(x^{10}y^2 + x^2y^{10}) + 15(x^8y^4 + x^4y^8) + 20x^6y^6.$$

Each parenthesized group is itself a symmetric polynomial in x, y — a *power-sum* of the form $p_{a,b}(x, y) := x^a y^b + x^b y^a$ (with the middle term $20x^6y^6$ being the $a = b$ case, where the two summands coincide). The Pascal coefficients $1, 6, 15, 20$ — *half* of row 6 — index the distinct pairs $\{12, 0\}, \{10, 2\}, \{8, 4\}, \{6, 6\}$.

1.8 Remark — A Polynomial Expression for r^{12}

In polar coordinates, $x = r \cos \theta$, $y = r \sin \theta$, and so

$$x^2 + y^2 = r^2 \implies (x^2 + y^2)^6 = r^{12}.$$

Thus the expansion gives a *polynomial* in (x, y) equal to r^{12} :

$$r^{12} = x^{12} + 6x^{10}y^2 + 15x^8y^4 + 20x^6y^6 + 15x^4y^8 + 6x^2y^{10} + y^{12}.$$

This is possible because 12 is even. For *odd* powers of r (e.g., $r^3 = (x^2 + y^2)^{3/2}$), no polynomial expression in (x, y) exists — r is not algebraic over $\mathbb{R}[x, y]$ once one moves outside even integer exponents.

The general fact: $r^{2m} = (x^2 + y^2)^m$ is a homogeneous polynomial of degree $2m$ in (x, y) for every non-negative integer m . Its coefficients are Pascal's row m , applied to the pairs $\{x^{2(m-k)}, y^{2k}\}$.

1.9 Remark — Why the Substitution Trick Works

Setting $u = x^2, v = y^2$ turns the problem into a *plain* binomial expansion $(u + v)^6$ — which we already know how to do. The trick is general: any binomial $(A + B)^n$ where A and B are themselves nicely-shaped expressions (powers, products, polynomials) can be expanded by Pascal's rule first, then expanded in the inner variables only if needed.

For *this* problem the inner expansions are trivial — $(x^2)^j = x^{2j}$ is one term, not many. So no second pass is required. For more involved problems like $(x^2 + y^3)^6$, one would also expand each $(x^2)^{6-k}(y^3)^k = x^{2(6-k)}y^{3k}$ in the same step. The general rule: *every* binomial-theorem expansion is one pass of Pascal's row, applied to *whatever* the two summands happen to be.